Calculus I Project

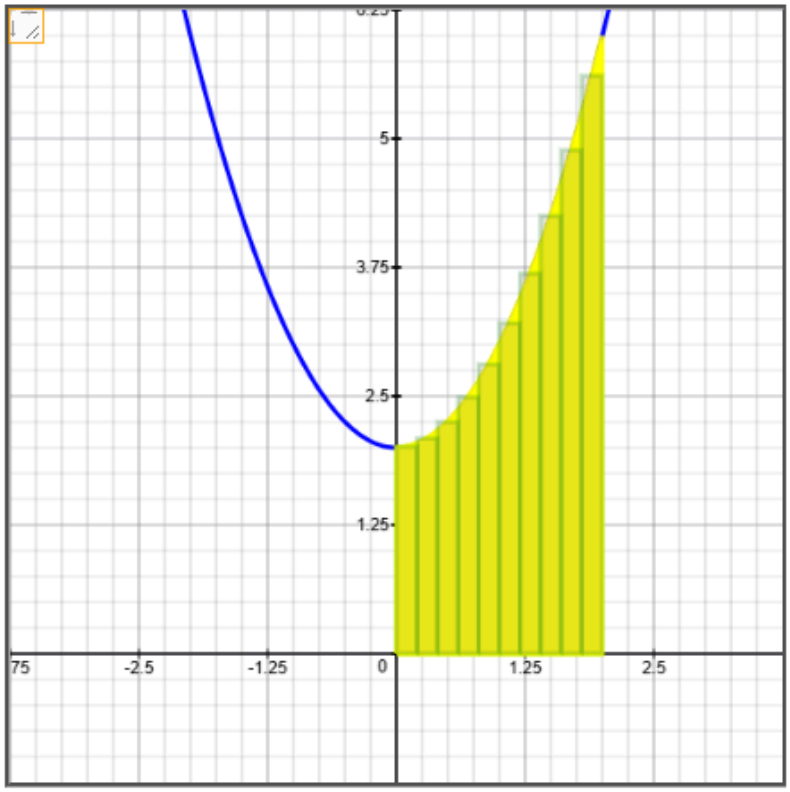
Objective: Find the exact area between the x-axis and $f(x) = x^2 + 2$ for $0 \le x \le 2$.

Part 1: Estimating Area Under Curve

Let $a = 0$, $b = 2$ and $\Delta x = \dfrac{b-a}{n} = \dfrac{2}{n}$, where n is the number of subintervals.

a) Use "Area Under Curve" program to find sum of areas of rectangles for $n = 10$: ___?___

Δx = width of each rectangle $= \dfrac{b-a}{n} = \dfrac{2}{10} = 0.2$



Note: Partitioning the interval $[0, 2]$ into 10 subintervals:

$a = 0 = x_0 \le x_1 \le x_2 \le x_3 \ldots \le x_8 \le x_9 \le x_{10} = 2 = b;$

$x_0 = a = 0$

$x_1 = a + 1 \cdot \Delta x = 0.2$

$x_2 = a + 2 \cdot \Delta x = 0.4$

$x_3 = a + 3 \cdot \Delta x = 0.6,$

$x_4 = a + 4 \cdot \Delta x = 0.8,$

$x_5 = a + 5 \cdot \Delta x = 1,$

$x_6 = a + 6 \cdot \Delta x = 1.2,$

$x_7 = a + 7 \cdot \Delta x = 1.4,$

$x_8 = a + 8 \cdot \Delta x = 1.6,$

$x_9 = a + 9 \cdot \Delta x = 1.5,$

$x_{10} = a + 10 \cdot \Delta x = 2.$

b) Use "Area Under Curve" program to find sum of areas of rectangles for $n = 20$: ?

Δx = width of each rectangle $= \dfrac{b-a}{n} = \dfrac{2}{20} = 0.01$

Note: Partitioning the interval $[0, 2]$ into 20 subintervals:

$a = 0 = x_0 \le x_1 \le x_2 \le x_3 \ldots \le x_{18} \le x_{19} \le x_{20} = 2 = b;$

$x_0 = a = 0$

$x_1 = a + 1 \cdot \Delta x = 0.1$

$x_2 = a + 2 \cdot \Delta x = 0.2$

$x_3 = a + 3 \cdot \Delta x = 0.3$

$x_4 = a + 4 \cdot \Delta x = 0.4$

...

$x_{15} = a + 15 \cdot \Delta x = 1.5,$

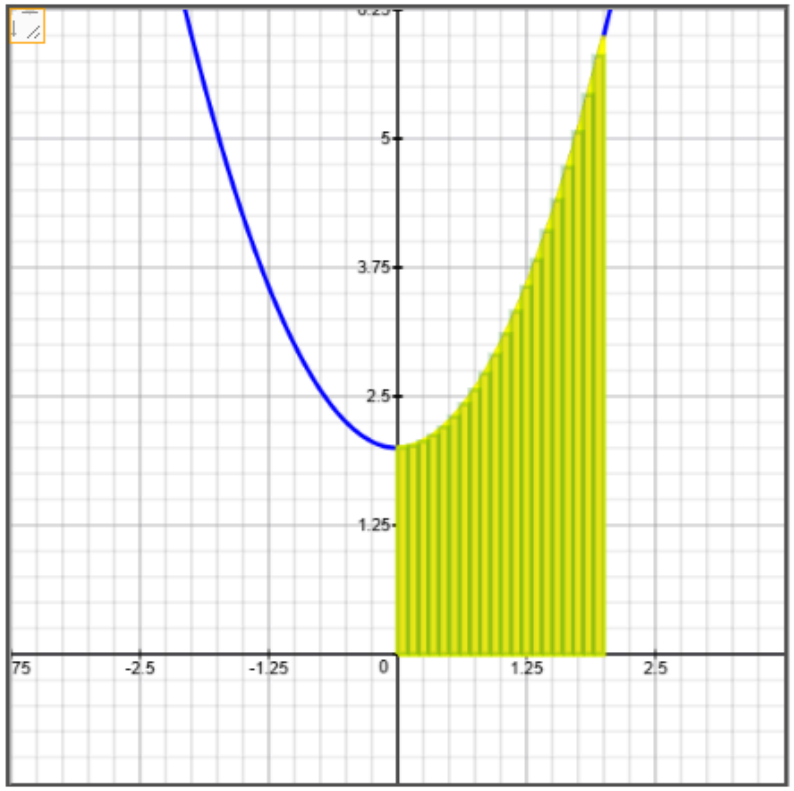
$x_{16} = a + 16 \cdot \Delta x = 1.6,$

$x_{17} = a + 17 \cdot \Delta x = 1.7,$

$x_{18} = a + 18 \cdot \Delta x = 1.8,$

$x_{19} = a + 19 \cdot \Delta x = 1.9,$

$x_{20} = a + 20 \cdot \Delta x = 2.$



c) Use "Area Under Curve" program to find sum of areas of rectangles for $n = 1{,}000$: ___?___

Δx = width of each rectangle $= \dfrac{b-a}{n} = \dfrac{2}{1000} = 0.002$

Note: Partitioning the interval $[0, 2]$ into 20 subintervals:

$a = 0 = x_0 \le x_1 \le x_2 \le x_3 \ldots \le x_{998} \le x_{999} \le x_{1000} = 2 = b;$

$x_0 = a = 0$

$x_1 = a + 1 \cdot \Delta x = 0.002$

$x_2 = a + 2 \cdot \Delta x = 0.004$

$x_3 = a + 3 \cdot \Delta x = 0.006$

$x_4 = a + 4 \cdot \Delta x = 0.008$

...

$x_{15} = a + 15 \cdot \Delta x = 1.990,$

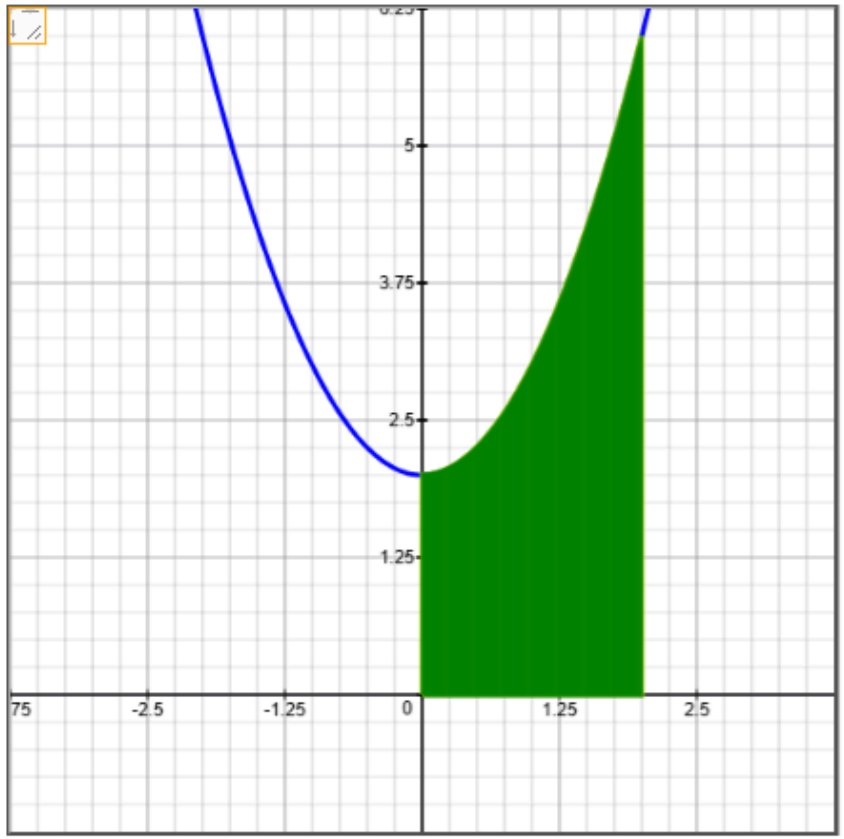
$x_{16} = a + 16 \cdot \Delta x = 1.992,$

$x_{17} = a + 17 \cdot \Delta x = 1.994,$

$x_{18} = a + 18 \cdot \Delta x = 1.996,$

$x_{19} = a + 19 \cdot \Delta x = 1.998,$

$x_{20} = a + 20 \cdot \Delta x = 2.$



Part 2: Finding Exact Area Under Curve

Partitioning the interval $[0, 2]$ into n subintervals: $a = 0 = x_0 \le x_1 \le x_2 \le x_3 \ldots \le x_{n-2} \le x_{n-1} \le x_n = 2 = b$

$x_0 = a = 0$

$x_1 = a + 1 \cdot \Delta x = 0 + 1 \cdot \left(\frac{2}{n}\right)$

$x_2 = a + 2\Delta x = 0 + 2 \cdot \left(\frac{2}{n}\right)$

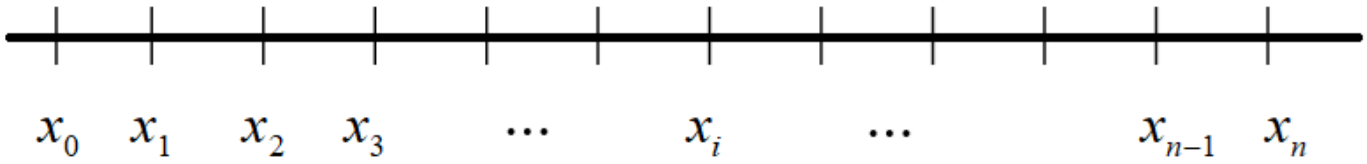
$x_3 = a + 3\Delta x = 0 + 3 \cdot \left(\frac{2}{n}\right)$

....

$x_i = a + i \cdot \Delta x = 0 + i \cdot \left(\frac{2}{n}\right)$

....

$x_n = n \cdot \Delta x = n \cdot \frac{2}{n} = 2$



The midpoint of each subinterval:

Subinterval 1: $\text{midpoint } x_1^m = a + 0.5 \cdot \Delta x = 0.5 \cdot \left(\frac{2}{n}\right)$

Subinterval 2: $\text{midpoint } x_2^m = a + 1.5 \cdot \Delta x = 1.5 \cdot \left(\frac{2}{n}\right)$

Subinterval 3: $\text{midpoint } x_3^m = a + 2.5 \cdot \Delta x = 2.5 \cdot \left(\frac{2}{n}\right)$

Subinterval 4: $\text{midpoint } x_4^m = a + 3.5 \cdot \Delta x = 3.5 \cdot \left(\frac{2}{n}\right)$

Subinterval *i*: $\text{midpoint } x_i^m = a + (i - 0.5) \cdot \Delta x = 0 + (i - 0.5) \cdot \left(\frac{2}{n}\right) = \frac{2}{n} \cdot i - \frac{1}{n}$

The height of each rectangle:

Height of Rectangle 1: $f(x_1^m) = f\left(0.5 \cdot \left(\frac{2}{n}\right)\right) = \left[0.5 \cdot \left(\frac{2}{n}\right)\right]^2 + 2$

Height of Rectangle 2: $f(x_2^m) = f\left(1.5 \cdot \left(\frac{2}{n}\right)\right) = \left[1.5 \cdot \left(\frac{2}{n}\right)\right]^2 + 2$

Height of Rectangle 3: $f(x_3^m) = f\left(2.5 \cdot \left(\frac{2}{n}\right)\right) = \left[2.5 \cdot \left(\frac{2}{n}\right)\right]^2 + 2$

Height of *i*th Rectangle: $= f(x_i^m) = f\left(\frac{2}{n} \cdot i - \frac{1}{n}\right) = \left(\frac{2}{n} \cdot i - \frac{1}{n}\right)^2 + 2 = \left(\frac{2}{n} \cdot i - \frac{1}{n}\right)\left(\frac{2}{n} \cdot i - \frac{1}{n}\right) + 2$

$$= \frac{4}{n^2} \cdot i^2 - \frac{4}{n^2} i + \frac{1}{n^2} + 2$$

The area of each rectangle:

Area of rectangle 1: $A_1 = \text{height} \cdot \text{width} = f(x_1^m) \cdot \Delta x$

Area of rectangle 2: $A_2 = \text{height} \cdot \text{width} = f(x_2^m) \cdot \Delta x$

Area of rectangle 3: $A_3 = \text{height} \cdot \text{width} = f(x_3^m) \cdot \Delta x$

Area of rectangle 4: $A_4 = \text{height} \cdot \text{width} = f(x_4^m) \cdot \Delta x$

Area of ith rectangle: $A_i = \text{height} \cdot \text{width} = f(x_i^m) \cdot \Delta x = \left[\frac{4}{n^2} \cdot i^2 - \frac{4}{n^2} i + \frac{1}{n^2} + 2 \right] \cdot \left(\frac{2}{n} \right) = \frac{8}{n^3} \cdot i^2 - \frac{8}{n^3} i + \frac{2}{n^3} + \frac{4}{n}$

Sum of areas of n rectangles $= A_1 + A_2 + A_3 + \ldots + A_i + \ldots + A_n = \sum_{i=1}^{n} A_i = \sum_{i=1}^{n} \left(\frac{8}{n^3} \cdot i^2 - \frac{8}{n^3} i + \frac{2}{n^3} + \frac{4}{n} \right)$

$$= \sum_{i=1}^{n} \left(\frac{8}{n^3} \cdot i^2 \right) - \sum_{i=1}^{n} \left(\frac{8}{n^3} i \right) + \sum_{i=1}^{n} \left(\frac{2}{n^3} \right) + \sum_{i=1}^{n} \left(\frac{4}{n} \right)$$

$$= \frac{8}{n^3} \cdot \sum_{i=1}^{n} \left(i^2 \right) - \frac{8}{n^3} \sum_{i=1}^{n} (i) + \sum_{i=1}^{n} \left(\frac{2}{n^3} \right) + \sum_{i=1}^{n} \left(\frac{4}{n} \right)$$

Summation Formulas

$\sum_{i=1}^{n} c = c \cdot n$ $\quad$ c is a constant

$\sum_{i=1}^{n} i = 1 + 2 + 3 + \ldots + n = \frac{n^2}{2} + \frac{n}{2}$

$\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$

$\sum_{i=1}^{n} i^3 = 1^3 + 2^3 + 3^3 + \ldots + n^3 = \frac{1}{4} n^4 + \frac{1}{2} n^3 + \frac{1}{4} n^2$

$\sum_{i=1}^{n} i^4 = 1^4 + 2^4 + 3^4 + \ldots + n^4 = \frac{1}{5} n^5 + \frac{1}{2} n^4 + \frac{1}{3} n^3 - \frac{1}{30} n$

Sum of areas of n rectangles $= \frac{8}{n^3} \cdot \sum_{i=1}^{n} \left(i^2\right) - \frac{8}{n^3} \sum_{i=1}^{n} (i) + \sum_{i=1}^{n} \left(\frac{2}{n^3}\right) + \sum_{i=1}^{n} \left(\frac{4}{n}\right) =$ _____?_____ .

Exact Area $= \lim_{n \to \infty} \sum_{i=1}^{n} \left(\frac{8}{n^3} \cdot i^2 - \frac{8}{n^3} i + \frac{2}{n^3} + \frac{4}{n} \right) =$ _____?_____ .

Part 3: Area Under Curve and Antiderivative

$f(x) = x^2 + 2$

$F(x)$ = antiderivative of $f(x)$ = ___?___

$\int_0^2 (x^2 + 2)\,dx = F(2) - F(0) =$ ___?___

How are the exact area in Part (2) and definite integral $\int_0^2 (x^2 + 2)\,dx$ related?

Part 4: Numerical Integration and Google Sheets

Trapezoid's Rule: area under curve $= \int_{a}^{b} f(x) \cdot dx$

$$\approx \frac{b-a}{n}\left[1 \cdot f(x_0) + 2 \cdot f(x_1) + 2 \cdot f(x_2) + \ldots + 2 \cdot f(x_{n-2}) + 2 \cdot f(x_{n-1}) + 1 \cdot f(x_n)\right]$$

$f(x) = x^2 + 2.$

Let $a = 0,\ b = 2,$ and $n = 100.$

Use Google Sheets to complete the following table:

i	Xi	Xi	f(Xi)	multiplier (m)	m·f(Xi)
0	x0	0	2	1	1
1	x1	0.02	2.0004	2	2.000099998
2	x2	0.04	2.0016	2	2.00039996
3	x3	0.06	2.0036	2	2.000899798
4	x4	0.08	2.0064	2	2.001599361
5	x5	0.1	2.01	2	2.002498439
6	x6	0.12	?	?	?
7	x7	0.14	?	?	?
8	x8	0.16	?	?	?
9	x9	0.18	?	?	?
10	x10	0.2	?	?	?
11	x11	0.22	?	?	?
12	x12	0.24	?	?	?
13	x13	0.26	?	?	?
14	x14	0.28	?	?	?
15	x15	0.3	?	?	?
16	x16	0.32	?	?	?
17	x17	0.34	?	?	?

18	x18	0.36	?	?	?
19	x19	0.38	?	?	?
20	x20	0.4	?	?	?
21	x21	0.42	?	?	?
22	x22	0.44	?	?	?
23	x23	0.46	?	?	?
24	x24	0.48	?	?	?
25	x25	0.5	?	?	?
26	x26	0.52	?	?	?
27	x27	0.54	?	?	?
28	x28	0.56	?	?	?
29	x29	0.58	?	?	?
30	x30	0.6	?	?	?
31	x31	0.62	?	?	?
32	x32	0.64	?	?	?
33	x33	0.66	?	?	?
34	x34	0.68	?	?	?
35	x35	0.7	?	?	?
36	x36	0.72	?	?	?
37	x37	0.74	?	?	?
38	x38	0.76	?	?	?
39	x39	0.78	?	?	?
40	x40	0.8	?	?	?
41	x41	0.82	?	?	?
42	x42	0.84	?	?	?

43	x43	0.86	?	?	?
44	x44	0.88	?	?	?
45	x45	0.9	?	?	?
46	x46	0.92	?	?	?
47	x47	0.94	?	?	?
48	x48	0.96	?	?	?
49	x49	0.98	?	?	?
50	x50	1	?	?	?
51	x51	1.02	?	?	?
52	x52	1.04	?	?	?
53	x53	1.06	?	?	?
54	x54	1.08	?	?	?
55	x55	1.1	?	?	?
56	x56	1.12	?	?	?
57	x57	1.14	?	?	?
58	x58	1.16	?	?	?
59	x59	1.18	?	?	?
60	x60	1.2	?	?	?
61	x61	1.22	?	?	?
62	x62	1.24	?	?	?
63	x63	1.26	?	?	?
64	x64	1.28	?	?	?
65	x65	1.3	?	?	?
66	x66	1.32	?	?	?
67	x67	1.34	?	?	?
68	x68	1.36	?	?	?

69	x69	1.38	?	?	?
70	x70	1.4	?	?	?
71	x71	1.42	?	?	?
72	x72	1.44	?	?	?
73	x73	1.46	?	?	?
74	x74	1.48	?	?	?
75	x75	1.5	?	?	?
76	x76	1.52	?	?	?
77	x77	1.54	?	?	?
78	x78	1.56	?	?	?
79	x79	1.58	?	?	?
80	x80	1.6	?	?	?
81	x81	1.62	?	?	?
82	x82	1.64	?	?	?
83	x83	1.66	?	?	?
84	x84	1.68	?	?	?
85	x85	1.7	?	?	?
86	x86	1.72	?	?	?
87	x87	1.74	?	?	?
88	x88	1.76	?	?	?
89	x89	1.78	?	?	?
90	x90	1.8	?	?	?
91	x91	1.82	?	?	?
92	x92	1.84	?	?	?
93	x93	1.86	?	?	?
94	x94	1.88	?	?	?

96	x96	1.92	?	?	?
97	x97	1.94	?	?	?
98	x98	1.96	?	?	?
99	x99	1.98	?	?	?
100	x100	2	?	?	?

Sum of values in last column = ___?___

Area Under Curve = $\int_a^b f(x) \cdot dx \approx$ ___?___

Part 5: Summary

Explain when the Fundamental Theorem of Calculus, $\int_a^b f(x) \cdot dx = F(b) - F(a)$, to find the exact area and when to use numerical integration method like Trapezoid's Rule.